LIBBY — PATIENT/CAREGIVER TRACK

aml-mds-related-rr-tp53-aberrant-hla-pending-x7q2

Plain-language summary for patient and caregiver

Generated 2026-07-06

This is decision-support information, not a treatment plan. Please share it with your oncologist before making any decisions based on what's here.

What this page is

Libby is an experimental tool that reads a person's cancer details and the published research, then lays out the treatment options worth talking through with the oncology team. This page is the plain-language version of that work. A separate, more technical page holds the same options written for clinicians.

One thing to know up front. This is not a simple case, and the honest answer to "what should I do" is "run a short set of tests first, because those results decide which doors are actually open." So this page leads with the tests, then walks through the treatments in the order they would most likely be used.

What we know about your cancer

You have acute myeloid leukemia, or AML, that grew out of an earlier bone-marrow condition (MDS). It has come back and is not responding to your most recent chemotherapy (the FLAG-IDA plus venetoclax combination).

Right now the leukemia is active. Your bone marrow is roughly 85% "blasts," which are the immature leukemia cells, so it has largely crowded out normal blood production. There is also a lump of leukemia in your right thigh. Doctors call that a myeloid sarcoma, which is just a solid collection of leukemia cells sitting outside the marrow.

The leukemia came back about six years after your first donor stem-cell transplant, which used a matched sibling as the donor. The way it returned suggests the leukemia found a way around the donor's immune cells rather than the transplant simply wearing off. That distinction matters, and it shapes a lot of what follows.

A few more pieces. The chromosome changes in the leukemia are complex and high-risk. The leukemia cells carry two surface markers, CD33 and CD123, that some treatments can lock onto. And a handful of results that the whole plan hinges on are not back yet: your tissue type (HLA), a pair of markers called HA-1 and HA-2 in both you and your sibling donor, and a gene called TP53 whose status is genuinely unsettled and is being confirmed rather than assumed.

What you told us matters most

You said your goal is a cure, and you are willing to accept harder, riskier treatment for a real shot at controlling the disease. You noted that this willingness should be looked at again once your fitness and heart tests are done, since those are still outstanding.

You did not rule out any type of treatment. A second transplant and immune-cell therapies are all acceptable to you. You are open to clinical trials, including traveling for one.

And the main reason you asked for this review: you want to know whether the immune-cell therapies, the ones that depend on your tissue type, are actually possible for you. That question sits at the center of everything below.

The first step everyone agreed on

Before any treatment, there is one step that is not a drug and that everyone lands on first: a short set of tests. They are the tests that unlock the options. None of them is toxic. They run on records that already exist, on blood and marrow samples you are already giving for disease checks, and on one sample from your sibling donor. Most results come back within one to three weeks.

Here is what they settle and why each one matters.

- **Your tissue type (HLA typing).** HLA is a set of markers on your cells that the immune system uses to tell "you" from "not you." One specific type, called HLA-A*02:01, is the master switch for most of the immune-cell therapies you came here to ask about. High-resolution typing was done around your first transplant, so the first move is to pull that result from the old records (often same-day). If it cannot be found, the test is repeated in about a week.
- **The HA-1 and HA-2 markers, in you and your sibling donor.** Even a fully matched sibling can differ from you at these smaller markers. That difference is what one of the most promising immune-cell therapies depends on, so both you and your donor need to be tested. This is a specialized test, usually run through a trial's central lab rather than an ordinary lab, and it takes one to two weeks.
- **Resolving your TP53 result.** A TP53 mutation was found back in 2019. A more recent test reported it as negative, but that test can miss low-level changes and certain kinds of gene loss, so it cannot be trusted to say the mutation is gone. The team is deliberately not calling this either way. Instead they are confirming it by reading the newer, more sensitive test alongside a protein stain, a chromosome test for the relevant region, and the original 2019 report. Whether a particular TP53 option is on the table depends on how this lands.
- **Whether the relapse is coming from your own cells.** After a donor transplant, a relapse can, rarely, come from the donor's cells instead of yours. A test called chimerism, run on the leukemia cells specifically, confirms which it is. This is load-bearing for two reasons: the immune-cell therapies are designed to aim donor cells at your cells, and if the leukemia turned out to be donor-derived, the whole plan would change.

While these results come in, your transplant team can hold parallel conversations with the trial programs so nothing waits longer than it has to.

The options

The plan that comes out of all this is a sequence, not a single drug. Think of it as three moves in order.

First, bring the leukemia in the marrow down. A second transplant cannot be done while the marrow is 85% leukemia, so the disease has to be reduced first. Second, a second donor stem-cell transplant, which is the only option here with a real chance at cure. Third, if the tissue-typing tests allow it, add immune-cell therapy on top to keep the leukemia from coming back.

A word on how a transplant fights leukemia, because most of the newer options rest on it. A donor transplant is not only a fresh set of blood cells. The donor's immune cells can recognize the leukemia as foreign and attack it. Doctors call that graft-versus-leukemia, sometimes shortened to GVL. The catch is that those same donor cells can also attack your healthy tissues, which is called graft-versus-host disease, or GVHD. Much of the newest work is about aiming the anti-leukemia effect more precisely so it hits the cancer while sparing the rest of you.

The options below are in the order they were ranked. Some are steps toward the transplant, some are the transplant itself, and some are the immune-cell therapies that only become possible depending on your test results.

Option 1 - A second donor stem-cell transplant

This is the backbone of the whole plan and the one option with a genuine cure fraction. In a large registry of second transplants, among patients most like you (a matched relative as donor), about 37 out of 100 were still alive two years later, compared with about 25 out of 100 across everyone. Your long first remission, roughly six years, sits on the favorable side of those odds.

The risks are real and serious. A second transplant carries a substantial chance of life-threatening complications from the transplant itself, plus GVHD and infection. It also cannot be done at 85% marrow blasts, so it depends on the earlier options reducing the leukemia first. And the conditioning intensity, the treatment given right before the new cells go in, cannot be chosen safely until your heart and organ-function tests are done. Those tests have not happened yet.

This matches what you told us: curative intent, no treatment type ruled out. The open question is honest. This leukemia already slipped past donor immune cells once, and its genetics are the kind that can escape a second time, so even a good first remission does not guarantee a lasting result. It stays the top option because nothing else here offers a real cure, but only once the marrow is reduced and your fitness is checked.

Option 2 - Targeted-radiation conditioning to make the transplant possible

One of the hardest problems in your case is that your marrow is packed with leukemia right now, which normally makes a transplant impossible. This option is a way around that. It is an antibody that carries radiation and homes in on blood and marrow cells (it targets a marker called CD45), delivering the radiation where the leukemia lives while sparing the rest of the body. It is the conditioning step, done right before transplant, and it can carry an active, high-blast marrow into the transplant.

This is the best-tested option on the page. In a randomized trial, where half the patients got this and half got usual care, about 17 out of 100 had a lasting remission with this approach compared with essentially none on usual care. Just as important, it did not pile on extra severe side effects: serious side effects were about equally common on both sides, near 60 out of 100, which is what these patients already face.

Two honest caveats. Overall survival in that trial did not clearly separate between the two groups, largely because many people in the comparison group crossed over and received this treatment too, which muddies the survival picture. And the trial enrolled people heading to their first transplant, so whether it is offered for a second transplant has to be checked, and the follow-on study is not open for enrollment yet. There is also a chance the marrow-aimed radiation does not reach the thigh lump, which may need its own directed radiation.

Option 3 - A CD33-targeted drug to reduce the leukemia (gemtuzumab)

This is the one option a treating team can simply prescribe today, without a trial. It is an antibody aimed at CD33, one of the markers confirmed on your leukemia, and it carries a cell-killing payload to those cells. In an older study, about 30 out of 100 patients went into remission. But those patients were less heavily treated than you are, so the real number for you is likely lower. This is a way to reduce the disease before transplant, not a cure on its own.

The main risk is liver toxicity, including a condition where small veins in the liver get blocked. That same liver risk can complicate the transplant this drug is meant to bridge you toward, so the timing between the two has to be handled carefully.

One honest point about how well it might work. The test confirmed CD33 is present on your leukemia, but not how much of it there is, and these drugs tend to work better when there is a lot of the marker. A test that measures the amount should guide whether this is worth using, rather than assuming it will help.

Option 4 - Extra donor immune cells (donor lymphocyte infusion)

This is a straightforward way to boost the graft-versus-leukemia effect: give you more of your sibling donor's immune cells. Because your donor is available, it is readily doable. Overall, adding these cells was linked to about 21 out of 100 patients alive at two years, versus about 9 out of 100 without. But in patients with active, high-burden disease like your marrow right now, the figure was closer to 15 out of 100.

So the honest read is that this works best after the leukemia has been reduced and alongside the transplant, not as a stand-alone treatment poured into a full marrow. Given into active disease, it mostly drives GVHD without much benefit. The main risk is GVHD, which could also complicate a later transplant. Your long first remission is exactly the feature that predicts these cells could help, but the timing has to be right.

Option 5 - HA-1/HA-2 immune-cell therapy (the reason for this review)

This option is only available if the tissue-typing tests come back the right way: HLA-A*02:01 positive, and a specific HA-1/HA-2 difference confirmed between you and your sibling donor. If either result does not line up, this option is off the table. And because this page only ranks options tied to your specific targetable features, a result that closes this door does not open a different one on this page. Standard care for the leukemia still exists, but that is a separate conversation with your treating team, not something this page tries to rank.

Here is what it is, when it is possible. This is the immune-cell therapy you came to ask about, and it has the highest ceiling of anything on the page. Doctors take your donor's immune cells and engineer them to target HA-1 or HA-2, which are minor markers found mainly on blood-lineage cells. Because those markers sit mostly on blood cells and not on other tissues, the therapy aims the anti-leukemia attack at the leukemia while largely sparing the organs that GVHD would otherwise hit. It is graft-versus-leukemia, pointed more precisely.

The evidence is early and promising in equal measure. In a small first-in-human study of nine patients whose leukemia had returned after a transplant, four of the nine went into or held remission, one for more than two years, with no dose-limiting side effects. That is a real signal, but it is a small, early study, and the detailed side-effect breakdown could not be retrieved from the published paper, so "no dose-limiting toxicities in nine patients" is most of what is known about safety. This therapy is given after the second transplant, so it does nothing for your marrow right now. It is the layer meant to keep the leukemia from coming back, and it hangs entirely on whether the HA-1/HA-2 test shows the right difference, which is close to a coin flip until the pair is actually tested.

Option 6 - A chemo-based way to deliver the transplant (FLAMSA-RIC)

This is the chemotherapy-based alternative to Option 2 for the same job: getting a marrow full of leukemia into a transplant. It combines sequential chemotherapy with a reduced-intensity transplant and a planned dose of donor immune cells afterward. In a study built for exactly the hard, high-risk situation you are in, about 88 out of 100 patients reached remission and about 42 out of 100 were alive at two years, and the high-risk subgroup did as well as lower-risk patients.

This is not really a separate treatment so much as a different route to the same transplant. Which route your center uses, this chemo-based one or the targeted-radiation one in Option 2, is a decision for the transplant team, guided partly by your heart and organ tests. The main risk is GVHD, including from the planned donor-cell dose.

Option 7 - A two-target immune-cell therapy (dual CD33/CD123 CAR-T)

This option carries caveats. Here is the tradeoff to weigh. The idea behind it is smart: your leukemia has already changed over time, so betting on a single target risks the leukemia dodging it. This therapy hits two markers at once, CD33 and CD123, both confirmed on your cells, to close off that escape route.

The problem is that there is no human data at all. Everything known about it comes from laboratory and animal studies, so nobody can tell you how well it works in people or how to manage its side effects, which are expected to include an inflammatory reaction and destruction of healthy blood-forming cells. On top of that, the trial sites are overseas, which raises real practical and coverage questions before it could even be considered. If a slot never materializes, the CD33-targeted drug in Option 3 is the available fallback for that target.

Option 8 - A CD123-targeted drug to reduce the leukemia (pivekimab)

This option carries caveats. It is another way to reduce the disease before transplant, aimed at the CD123 marker, and its appeal is that it does not reuse the chemotherapy your leukemia already shrugged off. About 21 out of 100 patients responded and about 17 out of 100 reached remission.

The tradeoff is the liver. This drug carries the strongest level of warning for the same liver-vein blockage mentioned earlier, and that is exactly the kind of harm that could jeopardize the transplant it is meant to bridge you toward. For that reason, the on-label CD33 drug in Option 3 is often preferred for the same reduce-the-disease job, at least until your organ-function tests are back. As with the CD33 drug, a test of how much CD123 is present should guide whether it is worth the risk.

Option 9 - WT1 immune-cell therapy (a backup immune path)

This option is only available if your HLA-A*02:01 test comes back positive. It exists as a backup: if you are A*02:01 positive but the HA-1/HA-2 difference in Option 5 does not line up, this keeps an immune-cell path open. If the tissue-typing does not support it, this option is off the table too, and, as before, a closed door here does not open a different ranked option on this page. That would be a conversation with your treating team about standard care.

When it is possible, this therapy uses engineered immune cells aimed at a protein called WT1 on the leukemia. In a small study of 12 patients, it was used to prevent relapse after a transplant, and all of them stayed relapse-free at about 44 months. That sounds striking, but read it carefully: those patients were already in remission and the therapy was preventing a return, not clearing a marrow that is 85% leukemia. The comparison group was not randomized, so the number is softer than it first appears. This fits best as a way to keep the leukemia from coming back after disease control, not as a treatment for active disease.

Option 10 - A TP53-targeted pill (rezatapopt)

This option is only available if your TP53 result is confirmed aberrant and the specific change is confirmed to be a particular variant called Y220C. That is a double condition, and your TP53 status is exactly the result that is still being sorted out, so this is doubly uncertain. If TP53 turns out not to be Y220C, this option is closed, and it does not open a different ranked option here.

When it is possible, it is a pill that reshapes that specific mutant protein back toward its normal working form. It is easier to tolerate than the transplant options, mostly low-grade nausea and some anemia. But the honest limits are large. Every response reported so far has been in solid tumors, not leukemia, at about 20 out of 100, and there is no leukemia data yet. It also works best paired with another drug, venetoclax, that your leukemia has already become resistant to. This sits off to the side of the main immune-cell plan, worth sorting out in parallel rather than counting on.

Questions to ask your oncologist

- Which of my tissue-typing and gene tests can be pulled from my first-transplant records today, and which need to be redone? When will we have all of them back?
- If the HLA and HA-1/HA-2 results do not line up for the immune-cell therapies, what does my plan look like then?
- What would it take to get my marrow leukemia low enough to make a second transplant possible, and how will we know we are there?
- Between the targeted-radiation conditioning and the chemo-based conditioning, which does this center offer, and which is safer for my heart and organs?
- What heart and organ-function tests do I need before a transplant, and could the results change whether transplant is an option at all?
- If we use a CD33- or CD123-targeted drug to reduce the leukemia first, how will you time it so the liver risk does not threaten the transplant?
- Does the lump in my thigh need its own radiation, separate from the transplant plan?
- Is my TP53 result settled yet, and does it change anything we do right now while we wait?

Sources

The recommendations on this page draw on the following published studies and clinical trials. The numbers cited above come from these references.

Published studies (PubMed IDs): 11432892, 16110027, 17909197, 23918951, 28860210, 29515236, 29661755, 31235963, 38423051, 38683966, 39298738, 39811143, 40608889, 41740031.

Clinical trials (ClinicalTrials.gov IDs): NCT02665065, NCT04156256, NCT05473910, NCT06561152, NCT06616636, NCT06704152, NCT07157514, NCT07645469.